Proprietary to General Electric Company

Detailed Calibration

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	2.5 hours	Not Applicable

2 Overview

This document provides the necessary steps to detailed calibrate the system.

Last Revised:	09 January, 2014

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Phantom Holder	1	-	2331933-2	-
Quality Assurance Phantom	1	-	2144715	-
Quality Assurance Phantom	1	-	5498268	-
Bubble Level	1	-	-	-

3.2 Safety



⚠ CAUTION

No personnel or patients should be in the area of the gantry while scanning without adequate radiation safety precautions being utilized.

3.3 Required Conditions

Condition	Reference	Effectivity
Collimator Calibration should be performed prior to Detailed Calibration.	-	-
Recommend procedure completion by GE Service trained personnel.	-	-
Gantry covers and scan window installed.	-	-

4 Procedure

4.1 Setup

1. Move table to longitudinal home position.
2. Attach phantom holder to the end of cradle.
3. Mount phantom to phantom holder.
4. Place bubble level on top of phantom, positioned along Z-axis. Adjust phantom holder back knob until phantom is level front to back.

4.2 Calibration

1. Click [CALIBRATION].
2. Click [DETAILED CALIBRATION]. Gantry Balance is performed.

   NOTE: All [KV], [SFOV], and [SLICE COLLIMATION] are forced on and not selectable.
3. Click [CONFIRM].
4. Click [OK] in *Attention* window.
5. Press [START] on GSCB when it flashes. Air Calibrations are performed.
6. Center phantom.

   NOTE: Setup and centering must be completed and calibration started in less than 30 minutes, otherwise required to start over.

   a. Turn on laser alignment lights.

   b. Move cradle into gantry until axial external alignment lights are aligned with phantom's axial reference mark.

   c. Adjust phantom holder side knob until sagittal external alignment light is aligned with phantom's sagittal reference mark.

   d. Elevate table until coronal external alignment light is aligned with phantom's coronal reference mark.

   e. Record the table/cradle position (height and axial) displayed on the control display panel.

   f. Elevate table 40mm and move cradle to compensate axial position until axial external alignment lights are aligned with phantom's axial reference mark.

   g. Move table out of gantry 70mm.

      NOTE: Axial external alignment lights should be in water area of phantom.

   h. Set external landmark, and the control display panel displays '**190**'.

   i. Move cradle into gantry until the control display panel displays '**0**'.

   j. Click [CONFIRM].

   k. Press [START] on GSCB when it flashes. Phantom Centering is performed.

   l. Adjust phantom position if results are out of specification range.

      NOTE: Phantom center specification is ± 1.0mm.

   m. Repeat steps [j] - [l] until phantom is centered.

   n. Click [CONTINUE] when phantom is centered.
7. Press [START] on GSCB when it flashes. Water Phantom Calibrations are performed.
8. Elevate table 40mm again.
9. Center phantom.

   a. Click [CONFIRM].

   b. Press [START] on GSCB when it flashes. Phantom Centering is performed.

   c. Adjust phantom position if results are out of specification range.

      NOTE: Phantom center specification is ± 1.0mm.

   d. Repeat steps [a] - [c] until phantom is centered.

   e. Press [CONTINUE] when phantom is centered.

10. Press [START] on GSCB when it flashes. Water Phantom Calibrations are performed.
11. Click [OK] in *Attention* window.
12. Click [QUIT] in *Scanner Utilities* window.

5 Finalization

1. *Fast Calibration*
2. Save System State

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